

## Topology 1 Exam

January 2026

Be neat, do not write too small, and leave one-inch margins.

**Show your work and support your statements. Use a separate sheet of paper for each problem. Each numbered problem is worth 10 points.**

1.
  - (a) Let  $f, g : X \rightarrow Y$  be continuous maps. Define what it means for  $f$  to be homotopic to  $g$ .
  - (b) Suppose that the continuous maps  $f, g : X \rightarrow Y$  are homotopic and that the continuous maps  $h, k : Y \rightarrow Z$  are homotopic. Show that  $h \circ f$  is homotopic to  $k \circ g$ .
2. If  $j : A \hookrightarrow X$  is the inclusion of a retract, then show that the induced map on fundamental groups is injective.
3.
  - (a) State the Seifert-van Kampen theorem.
  - (b) Determine the fundamental group of  $S^1 \vee S^1$ .
4. Prove that there are infinitely many distinct homotopy types of spaces with one 0-cell, one 1-cell, and one 2-cell.
5. Let  $\mathbb{RP}^2$  denote the real projective plane.
  - (a) Give the universal cover of  $\mathbb{RP}^2$  including the covering map.
  - (b) What is the group,  $G$ , of covering transformations of this cover?
  - (c) How does it act on the universal cover?
  - (d) Does  $\mathbb{RP}^2$  have a cover,  $p : E \rightarrow \mathbb{RP}^2$ , whose fundamental group,  $\pi_1(E)$ , is  $\mathbb{Z}/(3)$ ? Explain why or why not.
  - (e) Does  $\mathbb{RP}^2$  have a cover,  $p' : E' \rightarrow \mathbb{RP}^2$ , whose covering transformation group,  $G(E')$ , is  $\mathbb{Z}/(5)$ ? Explain why or why not.
6. Give the definition of a chain complex of abelian groups and its homology groups.
7. Let  $(C, d)$ , and  $(C', d')$  be chain complexes of abelian groups.
  - (a) Define what it means for  $f : (C, d) \rightarrow (C', d')$  to be a morphism of chain complexes, also called a chain map.
  - (b) For  $n \in \mathbb{Z}$  define the induced map  $H_n(f) : H_n(C, d) \rightarrow H_n(C', d')$ .
  - (c) Prove that  $H_n(f)$  is well defined.
8. For  $n \geq 0$ , let  $\Delta^n = \{(x_0, \dots, x_n) \in \mathbb{R}^{n+1} \mid x_i \geq 0, \sum_i x_i = 1\}$ .
  - (a) Let  $X$  be a topological space. Define the singular chain complex  $S_\bullet(X)$ .
  - (b) Let  $X$  be the one-point space  $*$ . Compute  $H_n(X)$  for all  $n$ .
9. Use the cellular chain complex to compute the homology of the Klein bottle.
10.
  - (a) State the Mayer-Vietoris sequence for reduced singular homology. You do not need to define the connecting homomorphism.
  - (b) Let  $n \geq 0$ . Compute the reduced singular homology groups  $\tilde{H}_k(S^n)$  for all  $k$ .